



Security Assessment



Item 24 Internal Report

October 2026

Prepared for ether.fi



Disclaimer

This document is intended for **internal** use only.



Initial Commit Hash	Final Commit Hash
364e4679	8e0dab35

Protocol Scope

- src/interfaces/IBackedCCIPBridge.sol
- src/migration/StockMigrationModule.sol
- src/modules/stock-wrap/StockWrapModule.sol
- src/oracle/ERC4626RatePriceFeed.sol
- src/stock-withdraw/StockBridgeWithdrawModule.sol
- src/top-up/TopUpDest.sol
- src/top-up/bridge/BackedCCIPBridgeAdapter.sol

Low Severity Issues

L-01: StockMigrationModule cannot migrate some opted-out safes

Severity: Low	Impact: Medium	Likelihood: Low
Files: src/migration/StockMigration Module.sol	Status: Fixed	

Description:

If a gateway safe has stand-in tokens supplied to Aave and `isLendOptedOut(safe)` is true, the module calls `cashModule.processLendOptOut(safe)`. That call reverts when the opt-out matured but the safe still has open borrows. `processLendOptOut` reverts with `HasOpenBorrows`.

The keeper will not be able to migrate these safes.

Recommendations:

Before calling `cashModule.processLendOptOut(safe)` make sure to properly check the revert conditions (including open borrows) so one stuck supplied position no longer blocks the safe's loose balance from migrating.

Customer Response: Fixed in commit [8e0dab35f](#)

Fix Review: Fixed

Informational Issues

I-01. Wrapper approvals are not cleared after the deposit

Description: Both functions approve the wrapper for the exact raw amount and then call deposit. Neither function sets the approval back to zero. Several areas of the codebase already explicitly reset prior approvals.

`StockWrapModule` approves the requested raw amount before calling `deposit`; `TopUpDest.wrapStock` does the same. The [Backed wrapper](#) rounds that amount down to shares and spends only the displayed value of those shares, which can be less than the approval. Both callers leave a residual wrapper allowance after a successful deposit. This is a stale approval to the configured wrapper, not a demonstrated way for an arbitrary caller to spend the funds.

Locations:

- `src/top-up/TopUpDest.sol:304`
- `src/modules/stock-wrap/StockWrapModule.sol:122`

Recommendation: After the deposit, reset the approval to 0.

Customer's response: Fixed in commit [8eOdab35f](#)

Fix review: Fixed

I-02. ERC-4626 rate dependency

Description: The oracle follows xStocks' [recommended valuation model](#) by combining the wrapper's convertToAssets rate with an independent USD price feed for the underlying xStock. However, the conversion-rate component is not independently verified: it depends on the current xStock multiplier maintained by Backed's authorized multiplier updater. An incorrect or compromised multiplier update would directly distort the wrapper's collateral valuation and could affect Aave borrowing capacity and liquidations.

Current wrappers are resistant to donation-based exchange-rate manipulation; this is an external privileged-multiplier trust assumption.

Recommendation: The team should be aware of the trust assumptions around using the vault based rates

Customer's response: Fixed in commit [8e0dab35f](#)

Fix Review: Fixed

I-03. Share rounding leaves raw xStock dust in protocol contracts

Description: [BackedCCIPBridgeAdapter](#) and [StockBridgeWithdrawModule](#) send a displayed raw token amount through Backed's bridge. Its `transferFrom` spends the full allowance but rounds the transfer down to whole underlying shares, so raw shares can remain in [TopUpFactory](#) or [StockBridgeWithdrawModule](#). [TopUpDest.wrapStock](#) can likewise leave a raw share after depositing its displayed balance into the wrapper. These residual shares do not reach the intended recipient through that operation; the factory's normal recovery function also rejects supported bridge tokens. ([Backed token share conversion](#), [TopUpFactory recovery](#))

Recommendation: Consider documenting the small rounding leftovers, so that safe owners are aware of the modules behaviour

Customer's response: Fixed in commit [8e0dab35f](#)

Fix Review: Fixed

I-04. TopUpDest records requested raw amounts rather than actual share movement

Description: `TopUpDest.deposit` credits `deposits[raw]` by the requested ERC-20 amount even when the rebasing token transfer moves fewer shares. `wrapStock` then subtracts up to the full displayed raw balance although the wrapper may consume fewer shares and leave raw dust behind. The raw deposit record can therefore overstate what arrived or fall to zero while raw shares remain. ([TopUpDest](#), [Backed token](#))

Recommendation: Check the before and after balance to calculate the actual amounts that left the contract

Customer's response: Fixed in commit [8e0dab35f](#)

Fix Review: Fixed



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